# The Sine Rule

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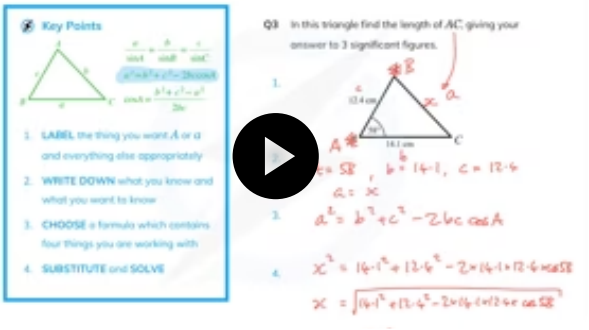


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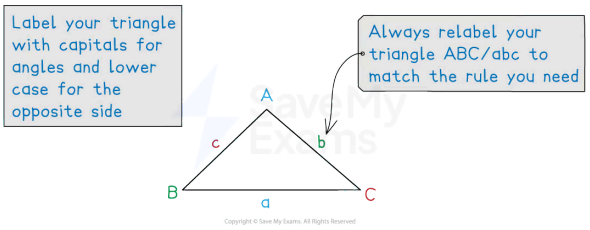
## Sine rule

### What is the sine rule?

- The **sine rule** is used in **non right-angled triangles**
  - It allows us to find missing side lengths or angles
- It states that for any triangle with angles $A$, $B$ and $C$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

- Where
  - $a$ is the side **opposite** angle $A$
  - $b$ is the side **opposite** angle $B$
  - $c$ is the side **opposite** angle $C$




Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

### How do I use the sine rule to find missing lengths?

        - *a* and *A*, or *b* and *B*, or c and C

- **Start by labelling your triangle** with the angles and sides
    - Angles have upper case letters
    - Sides **opposite** the angles have the equivalent lower case letter

- To find a **missing length**, substitute numbers into the formula

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

    - You only need to have two parts equal to each other (not all three)
        - Then solve to find the side you need

## How do I use the sine rule to find missing angles?

- To find a **missing angle**, it is easier to **rearrange** the formula first by flipping each part

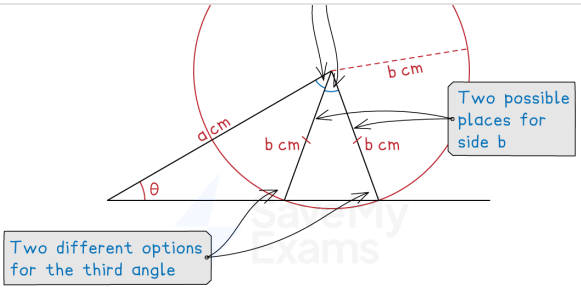
$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

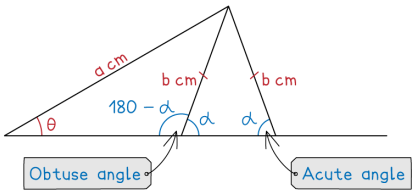
    - The angles are now in the numerators of the fractions
    - Substitute the values you have into the formula and solve
        - You will need to use **inverse sine** in your calculation, $\sin^{-1}(...)$

## What is the ambiguous case of the sine rule?

- Given information about a triangle, there may be **two different ways** to draw it
- In the diagram below, the lengths of two sides are given, *a* and *b*
    - A base angle is also given, $\theta$, but no angle near *b* is given
    - It turns out that there are **two possible ways** to arrange *b to* complete the triangle!
        - Both triangles have the correct values of *a*, *b* and $\theta$

- The other base angle could either be **obtuse or acute**
    - The **sine rule** only gives the **acute** answer on your calculator
        - You need to check the diagram to see if the angle you need is actually obtuse
        - If it is, use this rule: **obtuse angle = 180 – acute angle**

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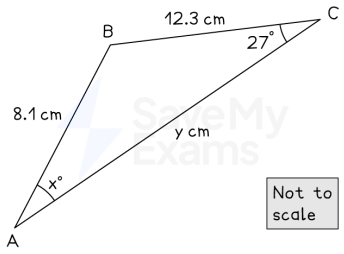
## Examiner Tips and Tricks

The sine rule is given in the formula booklet.

## Worked Example

The following diagram shows triangle ABC.

$AB = 8.1$ cm, $BC = 12.3$ cm and angle $BCA = 27°$.

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(a) Calculate the value of $x$.

**Answer:**

Label the sides of the diagram

$a \qquad b \qquad c$

In practice, you only need to equate two of these three parts

$$\frac{\sin x}{12.3} = \frac{\sin 27}{8.1}$$

$$\sin x = \frac{12.3 \sin 27}{8.1}$$

$$x = \sin^{-1}\left(\frac{12.3 \sin 27}{8.1}\right)$$

$$x = 43.58207...$$

$43.6°$ **(to 1 d.p)**

(b) Calculate the value of $y$ .

**Answer:**

To find y you need to know the angle opposite (angle ABC)
You know 27 and x from above, so subtract these from 180

$$\text{Angle } ABC = 180 - 27 - 43.58207...$$
$$= 109.41792...$$

y is a length so use the sin rule with the sides on the top

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$\frac{y}{\sin(109.41792...)} = \frac{8.1}{\sin 27}$$

$$y = \frac{8.1 \sin(109.41792...)}{\sin 27}$$

$$y = 16.82691...$$

$y = 16.8$ cm **(to 3 s.f.)**

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